Erdi Sayar | Curriculum Vitae

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Work Experience

- **Student Asistant**

RWTH Aachen University
10/2018 - 04/2019 *Aachen/Germany*

- **Research Asistant**

Izmir Institute of Technology
Teaching
- ME352 SYSTEM ANALYSIS AND CONTROL
- ME242 APPLIED MATHEMATICS FOR ENGINEERS

02/2017 - 02/2018 *Izmir/Turkey*

- **Research Asistant**

Turkish-German University
02/2016 - 02/2017 *Istanbul/Turkey*

Education

- **M.Sc**

RWTH Aachen University
Electrical Engineering, Information Technology and Computer Engineering
- Systems Engineering and Automation

2017 - 2020 *Aachen/Germany*
GPA : 1.5/1.00
Thesis : Anomaly detection in unknown environments using sensory agents
Advisors : Prof.Dr.Ascheid Gerd **Grade:** 1.0

- **Ph.D**

Izmir Institute of Technology
Mechanical Engineering
2016 - 2017 *Izmir/Turkey*

- **B.Sc**

Bochum University of Applied Sciences
Mechatronics Engineering
2013 - 2014 *Bochum/Germany*
GPA : 1.36/1.00
Thesis : Three phase asynchronous motor dynamic mathematical model and control Simulink
Advisors : Prof.Dr.-Ing Peter Brychta,Prof. Dr.-Ing. Arno Bergmann

- **B.Sc**

Kocaeli University
Mechatronics Engineering
2010 - 2015 *Kocaeli/Turkey*
GPA : 3.82/4.00
Valedictorian,Ranked first in the faculty of engineering.Graduated with top honors.

Engineering Skills

Matlab & Simulink | ROS | C & C++ | Python | MAPLE | Linux | Ansys | Solidworks | Autocad | Proteus | Visual Basic | PSPICE | Eagle | HTML & CSS & Javascript | LaTeX

Projects

- Using Cosimir Software detect objects colors and sort them with respect to colors [Robotic Lecture]

- Using Solidworks software , designing "Mechanical Press"mechanism and displacement, velocity and acceleration analysis [Theory of Machine Lecture]

- Fuzzy Logic Controller and PID Controller Design for Aircraft Pitch Control [Fuzzy Logic Controller Lecture]

- Designing State Space PI with Coupling and without Coupling , Optimal Control with Kalman Filter [Modern Control Lecture]

- Using Fanuc 6 Axes Robot and programming with Karel Programming language for pick and place task [Robotic Lecture]

- More Information and Projects in erdisayar.github.io

Achievements

- Ranked 1st in the Faculty of Engineering ,Kocaeli University (over 1000 students)

- Ranked 1st in Mechatronics Department ,Kocaeli University

- (TUBITAK)Turkish National Science Foundation 2211/A PhD scholarship

- Outstanding Teaching Assistant ,Izmir Institute of Technology

Publications & Certificates

- Fuzzy Logic Controller and PID Controller Design for Aircraft Pitch Control at IFFTOM Conference

- Assistive Handwriting Haptic Mechanism Using Deep Learning Speech Recognition at WCAIML Conference

- ROS Summer School Certificate at FH Aachen

Languages

Turkish Native
English TOEFL 82/120 ● ● ● ● ○
German Telc Deutsch Zertifikat ● ● ● ● ○

Area of Interest

•Robotic, Control, Machine Learning

Taking Lectures relating to Area of Interest

- Machine Learning $[R,E,M]$
- Artificial Intelligence $[R,E,M]$
- Machine Learning Laboratory $[R,E,M]$
- Advanced Robotic Kinematics And Dynamics $[R,E,M]$
- Multi Body Dynamic $[R,G,M]$
- Artificial Neural Network $[R,G,M]$
- Robotic Man-Machine Interaction 2 $[R,G,M]$
- Modern Control Systems $[R,E,M]$
- Rapid Control Prototyping $[R,G,M]$
- Multi Media Communication 2 $[R,E,M]$
- Principles of Robotics $[I,E,P]$
- Advanced Robotic Kinematics $[I,E,P]$
- Digitial Image Processing $[I,E,P]$
- Introduction to Fuzzy Logic $[K,E,B]$
- Robotics and Automation Systems $[K,T,B]$
- Fundamentals of Image Processing $[K,E,B]$
- Automatic Control $[K,T,B]$

$[X, Y, Z]$ *where* X *is name of university,* Y *is the instruction language,* Z *is the type of education.*
$X \in \{R, I, K\} \quad Y \in \{E, G, T\} \quad Z \in \{P, M, B\}$
$R = RWTH\ Aachen\ University, I = Izmir\ Institute\ of\ Technology,\ K = Kocaeli\ University$
$E = English,\ G = German,\ T = Turkish$
$P = Ph.D,\ M = Master,\ B = Bachelor$

Taking Lectures relating to Area of Interest

Machine Learning

Bayes Decision Theory,Probability Density Estimation Maximum Likelihood Approach,Kernel Methods,K-Means Clustering,Linear Discriminant Function,Logistic Discriminant Function, Least-Square Classification,Softmax Regression,Support Vector Machine,Adaboost,Neural Networks,Optimization Convolutional Neural Network,Recurrent Neural Networks.

Artificial Intelligence

Problem-Solving,Search and Games,Knowledge, Reasoning and Planning, Learning and Neural Network.

Machine Learning Laboratory

Supervised Learning,Feature Extraction,Unbiased Evaluation,Neural Networks, Backpropagation,Deep Learning,Segmentation and Optimization.

Advanced Robotic Kinematics And Dynamics

Orientation and Homogenous Transformation using Rotation Matrices ,Axis-Angle Representation,Quartenions,Direct and Inverse Kinematics, Forward and Inverse Differential Kinematics,Robot Dynamics with Lagrange and Newton-Euler, Identification Approach for Dynamic Parameters,Trajectory and Path Planning.

Multi Body Dynamic

Modeling and Kinematic, Equations of Motion and Linearization, Equation of States and Harmonic excitation.

Artificial Neural Network

Supervised and Unsupervised Learning,SVM , Self-organizing Maps and Associative Memory,Dynamics of Pulsneuron.

Robotic Man-Machine Interaction 2

Dynamic models of industrial Robots ,Machine Learning,Represantation and Processing of Knowledge,Fuzzy Logic,Newton-Euler and Lagrange.

Modern Control Systems

State Control, Optimal Control, Kalman Filter, Nonlinear Control, Lyapunov Theory.

Rapid Control Prototyping

Physical Modelling,Identification of Dynamic Systems,Discrete and hybrid simulation with Stateflow.

Multi Media Communication 2

Feature Extraction(Edge,Texture,Motion), Feature Transform and Classification.

Principles of Robotics

Spatial Forward and Inverse Kinematics,Velocity and Acceleration Analysis.

Advanced Robotic Kinematics

Kinematic and Dynamic Control of Serial Robotis.

Digitial Image Processing

Image Enhancement, Segmentation,Wavelet Transform, Color Image Processing,Morphology, ImageCompression, Camera Models.

Introduction to Fuzzy Logic

Fuzzification and Defuzzification methods,Fuzzy Logic Controller.

Robotics and Automation Systems

Resolving Langrange-Euler and Newton-Euler equations for robots,Developing the trajectory of robots.,Calculating Jacobian matrix.

Fundamentals of Image Processing

Spatial and Frequency domain filtering,Basic morphological operations,Basic segmentation methods.

Automatic Control

Representing the control systems based on their transfer functions and state space,Explaining the controllability and observability concepts,Recognizing the response of control systems at time and frequency domain,operation principles of PID algorithms and their industrial applications,Realizing the simulation of control systems at Matlab/Simulink environment.